

Appendix B

Rationales for the Identified Research Gaps

1. Reproducibility Issues:

The review found that important methodological details were not consistently reported in the reviewed studies. In several cases, information required to understand or reproduce the proposed approaches was unavailable in the published papers or associated repositories, requiring the researcher to contact the corresponding authors for clarification. This lack of comprehensive methodological reporting reduces transparency and hinders the reproducibility, independent validation, and fair comparison of related work generation approaches. Future research should therefore improve methodological reporting by providing sufficient implementation details, experimental settings, and supporting resources to enable reproducibility and facilitate independent validation and comparison of related work generation approaches.

2. Task Formulation:

The reviewed studies formulate related work generation by representing the target paper using its title, abstract, or full text to guide the generation process. However, this formulation does not explicitly represent how researchers develop literature reviews before writing

begins. Specifically, it does not model what should be discussed, how the review should be organised, or what writing style should be adopted. As discussed in Section 2.1.3 (Literature Review as an Argumentative Process), researchers first determine what should be discussed by identifying the relevant studies, major debates, research gaps, and the intended contribution of the current study. Section 2.1.4 (Content Expectations in Literature Review) shows how the review should be organised by establishing the background context, providing critical evaluation and meaningful comparison, and presenting the author’s contribution within a coherent structure. Finally, Section 2.1.6 (Styles of Literature Reviews and Transformation of Source Information) distinguishes what writing style the review should follow by identifying descriptive and integrative literature reviews. These aspects of literature review development are not explicitly represented in existing task formulations, which instead rely primarily on the textual representation of the target manuscript to guide generation. Furthermore, existing task formulations generally assume that the relevant reference papers have already been identified and are provided as input to the generation model. As discussed in Section 2.2.7 (Automatic Evaluation), generated related work sections are typically evaluated by comparing them with the published related work section, which serves as the gold standard. Although this provides a useful starting point for evaluation, it does not reflect the actual literature review writing process. Human researchers do not write related work sections from a gold standard. Instead, they retrieve, select, and analyse relevant papers before writing. Moreover, a single gold standard cannot fully represent the task, since multiple valid related work sections may be produced depending on what should be discussed, how the review should be organised, and what writing style should be adopted. Consequently, existing task formulations provide limited support for end-to-end related work generation.

3. Approaches:

The reviewed studies propose a range of approaches for automatic related work generation. Despite their methodological differences, a synthesis of the literature reveals common conceptual limitations. In particular, the reviewed approaches are primarily formulated from

a natural language processing perspective and provide limited support for the diversity of literature review writing scenarios encountered across different research domains. The following research gaps summarise these limitations. Although the reviewed approaches differ in their implementation, they formulate related work generation primarily as a natural language processing task. Several approaches attempt to incorporate characteristics of human-written related work sections, but these characteristics are primarily modelled to improve the performance of the natural language processing task. However, they do not explicitly draw upon the broader literature review writing process described in Section 2.1, where researchers determine what should be discussed, how the review should be organised, and what writing style should be adopted before writing. **As a result, existing approaches focus on improving the generation pipeline rather than modelling the literature review writing process itself.** Moreover, the reviewed approaches are generally developed and evaluated within a specific research setting, with limited consideration of how the generation process should adapt to different research domains. Section 9.5.1 suggests that the literature review writing process may vary according to the characteristics of the research domain, including the maturity of the topic, the availability of relevant literature, and the objectives of the review. These factors may influence what should be discussed, how the review should be organised, and what writing style should be adopted. Therefore, existing approaches provide limited support for adapting the generation process to different literature review writing scenarios.

4. Datasets and Experimental Validation:

The reviewed studies employ different datasets to develop and evaluate related work generation approaches. Despite recent progress, the literature reveals limitations related to benchmark datasets, domain coverage, and the interpretation of experimental results. The following research gaps summarise these limitations. Although several reviewed studies employ OARelatedWork, the literature continues to use different datasets with varying characteristics and evaluation settings. This limits the comparability of experimental results

across studies and makes it more difficult to assess progress in the field. Furthermore, the reviewed datasets are predominantly drawn from the computer science domain, providing limited evidence regarding the applicability of existing approaches to other research domains with different publication practices, literature characteristics, and related work writing conventions. The ablation studies reported in the reviewed literature provide limited insight into the extent to which the performance of LLM-based approaches is influenced by prior knowledge acquired during pre-training. Although some studies attempt to avoid data contamination, it remains unclear whether the reference papers used for related work generation were included in the LLM’s training data. Likewise, when relevant reference papers are unavailable to the LLM, the impact on the quality of the generated related work sections has received limited investigation. Consequently, the effect of pre-training knowledge and reference paper availability on the validity and interpretation of experimental results remains insufficiently understood.

5. User Interaction:

Among the reviewed approaches, only LitLLM provides limited pre-generation interaction, allowing users to influence the generation process before writing begins. In contrast, the reviewed studies provide little consideration of post-generation interaction. This is an important limitation because literature review writing is inherently iterative. Human researchers typically review, revise, and refine multiple versions of a related work section before producing the final manuscript, adjusting both the content and its organisation throughout the writing process. Existing approaches therefore provide limited support for human involvement across the iterative development of related work sections.

6. Automatic Evaluation:

The reviewed studies employ different automatic evaluation methods. However, a synthesis of the literature reveals limitations related to baseline comparisons and the use of LLMs as judges. When baselines belong to the Related Work Generation category described in Section 2.2.7, meaningful comparison requires alignment in task formulation, inputs, objec-

tives, and evaluation settings. Where these differ, it becomes difficult to determine whether performance differences reflect methodological improvements or differences between the underlying approaches. Furthermore, several reviewed studies employ LLMs as judges to assess the quality of generated related work sections. However, the prompts, implementation details, source code, and adaptation of the evaluation criteria are often insufficiently reported, limiting the reproducibility of LLM-based evaluation. The reviewed studies also provide limited justification for how the evaluation criteria are adapted to the task of related work generation. For example, Liu et al. (2025) assess relevance, but the definition of this criterion and the descriptions of some rating categories provide limited guidance on how relevance should be interpreted in the context of related work generation. Table 2.10 provides a synthesis of the quality criteria reported in the reviewed studies and offers a starting point for developing task-specific evaluation criteria. However, further research is needed to define and operationalise these criteria for automatic evaluation of related work generation.

7. Human Evaluation:

Only seven of the eleven reviewed studies conducted human evaluation, and the reported evaluations were generally limited in scope. A synthesis of these studies reveals inconsistencies in evaluation protocols, limited evidence about evaluator expertise, and limited qualitative analysis. The reviewed studies adopted different evaluation criteria and assessment methods. These differences make it difficult to compare findings across studies and to establish consistent evidence regarding the quality of generated related work sections. Several studies describe PhD students as domain experts. Although PhD students may possess substantial knowledge of their research topics, topic expertise does not necessarily imply extensive experience in literature review writing or broader aspects of scientific writing. The reviewed studies provide limited information about these additional forms of expertise, making it difficult to assess the suitability of evaluators for human evaluation. The reviewed studies also provide limited qualitative analysis of participants' feedback. Such analysis is needed to understand users' needs, identify weaknesses in the generation pipeline, and explain

why particular outputs are judged to be effective or inadequate. Although human studies are difficult and resource-intensive to conduct, they remain necessary for understanding the quality of generated content and improving support for literature review writing.

8. Scalability and Computational Constraints:

Some reviewed studies report scalability and computational constraints as limitations of their approaches. These include the computational cost of processing long documents, increased memory requirements, and the limited context windows of LLMs when handling large collections of reference papers. Such constraints may reduce the practicality of existing approaches for generating related work sections from large and diverse bodies of literature. Further research is therefore needed to develop approaches that maintain generation quality while improving computational efficiency and scalability.

9. Ethical Considerations:

Table 2.12 highlights that ethical considerations extend beyond technical performance to include issues such as transparency, bias, attribution, academic integrity, and responsible use. Although these considerations are recognised in the reviewed studies, they are generally discussed as individual concerns rather than as an integrated framework for the design and evaluation of related work generation systems. This suggests that future research should consider ethical considerations throughout the development lifecycle, from system design to evaluation and deployment.